

A Novel Spectral Slope and Intercept Parameterization of Power Spectral Density for Automated Neonatal Seizure Detection

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Abstract

Neonatal seizures represent the most prevalent neurological emergency in the neonatal intensive care unit (NICU), with 60–80% occurring subclinically without overt physical manifestations. Continuous multi-channel electroencephalography (EEG) is the clinical gold standard, but expert visual interpretation is labour-intensive, non-scalable, and prone to substantial inter-observer discordance. Existing automated detection frameworks predominantly rely either on high-dimensional heuristics that lack physiological interpretability or on end-to-end black-box deep neural networks that require vast amounts of labelled data and risk patient-level overfitting. In this work, we propose an ultra-compact neurophysiological biomarker for neonatal seizure identification based on a novel first-order parametric model of the log-transformed power spectral density (PSD). By fitting linear models to log-power versus log-frequency across canonical EEG frequency bands (δ : 0.5–4 Hz, θ : 4–8 Hz, α : 8–13 Hz, β : 13–30 Hz), we extract three physically grounded parameters per band: spectral roll-off slope (α), power axis intercept (β_0), and midband-predicted power (P_{mid}), yielding an efficient 12-dimensional descriptor per channel. We validate our proposed feature set on the benchmark Helsinki neonatal EEG corpus (79 subjects, 39 with multi-expert consensus annotations) across an extensive suite of eight systematic validation studies under strict 10-trial patient-level cross-validation. In baseline comparison studies using a linear classifier, our spectral slope parameterization achieves the highest test AUROC (0.615 ± 0.163) among all evaluated feature representations, outperforming Empirical Mode Decomposition (0.526 ± 0.159), Discrete Wavelet Transforms (0.474 ± 0.089), Band Power (0.506 ± 0.098), Hjorth parameters (0.560 ± 0.094), and Spectral Entropy (0.595 ± 0.146). Fisher discriminant ratios and Cohen’s d analyses confirm that delta-band slope ($J = 0.041$, $d = -0.261$) and midband power ($J = 0.054$, $d = +0.306$) drive strong class separation. When integrated into a deep feedforward neural network with moving-average temporal post-processing, the parameterization reaches a peak test AUROC of 0.858 and an F1-score of 0.731. These findings establish log-PSD linear parameterization as a computationally lightweight, clinically interpretable, and

highly discriminative biomarker for automated neonatal seizure monitoring.

Keywords: Neonatal seizures, Electroencephalography, Power spectral density, Spectral slope, Empirical mode decomposition, Biomarkers, Machine learning.

1 Introduction

Neonatal seizures are the most frequent neurological emergency encountered in the neonatal intensive care unit (NICU), signifying acute brain injury or systemic metabolic derangement [1, 2]. The estimated incidence is 1–5 per 1,000 live births among term infants, escalating dramatically to 57–132 per 1,000 in very-low-birth-weight preterm neonates [1, 3]. Common underlying aetiologies include hypoxic-ischaemic encephalopathy (HIE), intracranial haemorrhage, cerebral infarction, central nervous system infections, and inborn errors of metabolism [4]. Crucially, recurrent neonatal seizures exacerbate secondary metabolic stress, increase cerebral metabolic demand, and are strongly associated with long-term neurological sequelae, including post-neonatal epilepsy, cerebral palsy, cognitive deficits, and neurodevelopmental delay [4, 5]. Prompt and reliable detection is therefore paramount to guide targeted neuroprotective therapy.

Continuous multi-channel electroencephalography (EEG) remains the uncontested gold standard for diagnosing neonatal seizures [2, 6]. Bedside clinical assessment is profoundly inadequate: approximately 60–80% of all neonatal seizures exhibit no observable motor manifestations (electrographic-only seizures), while many abnormal neonatal movements mimic seizures without corresponding electrographic discharge (electro-clinical dissociation) [7]. Despite its diagnostic necessity, continuous EEG interpretation requires round-the-clock neurophysiologists. Such specialized coverage is scarce or unavailable in most NICUs worldwide. Furthermore, neonatal EEG morphology is uniquely challenging, marked by background discontinuity, tracé alternant patterns, age-dependent non-stationarities, and pervasive clinical artifacts [8]. Consequently, inter-observer agreement among expert neurologists remains only moderate ($\kappa \approx 0.40$ – 0.60), emphasizing the pressing clinical demand for objective, automated seizure monitoring systems [8, 9].

Prior approaches to automated detection have broadly followed two paradigms. The first paradigm relies on extracting vast batteries of hand-crafted features spanning the time, frequency, and time–frequency domains [10, 11]. Time-domain features, such as amplitude variance, zero-crossing rates, and non-linear complexity metrics, attempt to track waveform morphology [10]. Frequency-domain approaches conventionally integrate the power spectral density (PSD) over predefined bands [11]. Time–frequency methods, such as the Discrete Wavelet Transform (DWT) and Empirical Mode Decomposition (EMD), decompose the non-stationary signal into multiscale sub-bands or Intrinsic Mode Functions (IMFs), computing statistical moments across modes [12–15]. While these multi-domain feature vectors have demonstrated competitive performance when paired with classifiers like Support Vector Machines (SVMs) [11], they frequently encompass dozens to hundreds of redundant features, increasing computational burden and obscuring the underlying neurophysiological mechanisms.

The second paradigm involves end-to-end deep learning architectures, such as Convolutional Neural Networks (CNNs) and recurrent models, that learn representations directly from raw multi-channel traces [16–18]. While deep networks achieve high benchmark numbers under specific conditions, they behave as opaque black boxes, lack clinical interpretability, and exhibit vulnerability to patient-level distribution shifts when trained on modest clinical cohorts.

In this paper, we address the fundamental question of feature representation: *What is the most compact, physically meaningful parameterization of the EEG spectrum that captures neonatal seizure dynamics?*

We propose a novel neurophysiological biomarker based on a first-order parametric model of the log-transformed power spectral density. Rather than treating frequency bins independently or integrating coarse band power, we fit linear regression models to the log-power versus log-frequency spectrum across canonical bands (δ , θ , α , β). This parameterization extracts three physically grounded descriptors per band: the spectral roll-off slope (α), the power-axis intercept (β_0), and the midband-predicted power (P_{mid}), yielding an ultra-compact 12-dimensional feature vector per channel. This formulation is inspired by spectral line-fitting principles originally demonstrated by Sinha et al. [19] in photoacoustic tissue characterization, and connects directly to contemporary neuroscience regarding $1/f^x$ neural electrophysiological dynamics and excitation/inhibition balance [20–22].

Through eight comprehensive empirical studies on the open-access Helsinki neonatal EEG dataset [23], we demonstrate that this simple 12-dimensional parameterization carries superior discriminative separability over complex baseline representations (EMD, DWT, Hjorth, Spectral Entropy, and raw Band Power), captures millisecond-level seizure onset trajectories, and achieves robust di-

agnostic performance when deployed in a deep neural network pipeline.

2 Materials and Dataset

2.1 Helsinki Neonatal EEG Corpus

We evaluate our methods using the publicly accessible Helsinki University Hospital neonatal EEG database [23]. The corpus comprises continuous multi-channel scalp EEG recordings from 79 full-term neonates admitted to the NICU with suspected seizures or hypoxic–ischaemic encephalopathy. The median recording duration is 74 minutes per neonate (totaling over 100 hours of continuous data).

Seizure annotations were independently provided by three expert clinical neurophysiologists who reviewed the recordings in their entirety [23]. Because inter-observer disagreement is substantial in neonatal EEG [8], we adopt a strict **unanimous consensus protocol**: an epoch is assigned a seizure ground-truth label ($y = 1$) if and only if all three expert annotators independently confirmed seizure activity at that exact second. Epochs with discordant annotations or uncertain labels are purged to ensure a noise-free, gold-standard benchmark. Under this criterion, 39 of the 79 neonates exhibit verified seizure events, while the remaining 40 subjects represent seizure-free clinical controls.

2.2 Bipolar Montage and Epoching

Raw EEG data, originally sampled at 256 Hz, were re-referenced to a standardized 18-channel modified anterior–posterior bipolar (double banana) montage conforming to the international 10–20 system [6]:

- **Right Parasagittal:** Fp2–F4, F4–C4, C4–P4, P4–O2
- **Left Parasagittal:** Fp1–F3, F3–C3, C3–P3, P3–O1
- **Right Temporal:** Fp2–F8, F8–T4, T4–T6, T6–O2
- **Left Temporal:** Fp1–F7, F7–T3, T3–T5, T5–O1
- **Midline:** Fz–Cz, Cz–Pz

Continuous recordings were segmented into fixed, non-overlapping epochs of 1-second duration (256 samples per channel). The 1-second resolution provides fine-grained temporal fidelity, ensuring minimal latency for bedside clinical alerting.

2.3 Patient-Level Cross-Validation

A critical pitfall in automated EEG literature is intra-patient data leakage, where epochs from the same neonate appear in both training and testing folds, artificially inflating reported performance. To guarantee clinical validity, we enforce a strict **patient-level 10-trial cross-validation** scheme:

- **Training Set:** 28 neonates ($\sim 70\%$)
- **Validation Set:** 6 neonates ($\sim 15\%$)

- **Test Set:** 5 neonates (~15%)

Across all 10 independent trials, no patient’s data ever overlaps between train, validation, and test partitions. All preprocessing scalers, imputation parameters, and feature transformations are fitted strictly on the training partition and applied out-of-sample to validation and test folds.

3 Methodology

3.1 The Proposed Novel Feature Set: Log-PSD Linear Parameterization

The foundational hypothesis of this work is that neonatal electrographic seizures fundamentally alter the broadband power-law scaling ($1/f^\chi$) and localized oscillatory geometry of cortical electrophysiology [20, 21]. Rather than decomposing signals into dozens of arbitrary bins, we capture these spectral shifts via a parametric model of the log-transformed power spectral density.

3.1.1 Mathematical Formulation

For each 1-second epoch $x(t)$ ($N = 256$ samples at $f_s = 256$ Hz) on a given bipolar derivation, we estimate the one-sided power spectral density $P(f)$ using Welch’s modified periodogram with a Hann window:

$$P(f) = \frac{1}{MU} \left| \sum_{n=0}^{N-1} x[n]w[n]e^{-j2\pi fn/f_s} \right|^2 \quad (1)$$

where $w[n]$ is the window function and $U = \frac{1}{N} \sum w^2[n]$ is the window energy normalization factor.

We partition the spectrum into four canonical neurophysiological frequency bands:

- **Delta (δ):** 0.5 – 4.0 Hz
- **Theta (θ):** 4.0 – 8.0 Hz
- **Alpha (α):** 8.0 – 13.0 Hz
- **Beta (β):** 13.0 – 30.0 Hz

For each frequency band $b \in \{\delta, \theta, \alpha, \beta\}$, let $\{f_k\}_{k=1}^{K_b}$ denote the discrete frequency bins within the band interval $[f_{\text{low}}^{(b)}, f_{\text{high}}^{(b)}]$, and let $P(f_k)$ be the corresponding estimated power spectral densities.

We apply a natural logarithmic transformation to both frequency and power coordinates:

$$\xi_k = \ln(f_k) \quad (2)$$

$$\eta_k = \ln(P(f_k)) \quad (3)$$

We then formulate an ordinary least-squares (OLS) linear regression model over the log-log coordinates:

$$\eta_k = \alpha_b \xi_k + \beta_{0,b} + \epsilon_k, \quad k = 1, \dots, K_b \quad (4)$$

where ϵ_k represents the residual fitting error. The closed-form analytical solutions for the regression slope α_b and

intercept $\beta_{0,b}$ are:

$$\alpha_b = \frac{\sum_{k=1}^{K_b} (\xi_k - \bar{\xi})(\eta_k - \bar{\eta})}{\sum_{k=1}^{K_b} (\xi_k - \bar{\xi})^2} \quad (5)$$

$$\beta_{0,b} = \bar{\eta} - \alpha_b \bar{\xi} \quad (6)$$

where $\bar{\xi} = \frac{1}{K_b} \sum \xi_k$ and $\bar{\eta} = \frac{1}{K_b} \sum \eta_k$.

In addition to the slope and intercept, we extract the model-predicted power at the geometric center frequency of the band, denoted as the **midband power** ($P_{\text{mid},b}$):

$$f_{\text{mid}}^{(b)} = \sqrt{f_{\text{low}}^{(b)} \cdot f_{\text{high}}^{(b)}} \quad (7)$$

$$P_{\text{mid},b} = \alpha_b \ln(f_{\text{mid}}^{(b)}) + \beta_{0,b} \quad (8)$$

For each channel and each frequency band b , the extraction produces exactly three descriptors:

$$\theta_b = [\alpha_b, \beta_{0,b}, P_{\text{mid},b}]^T \in \mathbb{R}^3 \quad (9)$$

Aggregating across the four bands yields 12 features per channel:

$$\theta_{\text{channel}} = [\theta_\delta^T, \theta_\theta^T, \theta_\alpha^T, \theta_\beta^T]^T \in \mathbb{R}^{12} \quad (10)$$

For a multi-channel recording of $C = 18$ channels, the full epoch representation is $\mathbb{R}^{216}$ (or an average across channels $\mathbb{R}^{12}$).

3.1.2 Electrophysiological Rationale

The three parameters capture distinct, complementary facets of the neural power spectrum:

1. **Spectral Slope (α):** Normal background neonatal EEG exhibits steep pink-noise roll-off ($P(f) \propto 1/f^\chi$ with $\chi \approx 1-2$, yielding negative slopes $\alpha < 0$). During electrographic seizures, hyper-synchronous neuronal firing introduces narrowband rhythmic sinusoidal oscillations and sharp spike-wave discharges. In low frequencies, this induces flattening or pronounced steepening of local spectral decay, serving as a direct marker of altered excitation/inhibition balance [21].
2. **Power Intercept (β_0):** Reflects the broadband amplitude baseline. Paroxysmal seizure bursts dramatically elevate overall field potential variance, shifting the intercept upward.
3. **Midband Power (P_{mid}):** Provides a smoothed, robust measure of centralized resonant band power that is resilient to bin-level spectral noise or notch-filter edge effects.

3.2 Baseline Feature Extraction Paradigms

To benchmark the novel log-PSD feature set, we implemented six canonical feature extraction paradigms under identical preprocessing and evaluation conditions:

3.2.1 1. Empirical Mode Decomposition (EMD)

Following Huang et al. [14], EMD decomposes non-linear and non-stationary time series into Intrinsic Mode Functions (IMFs). For each 1-second epoch, signals are adaptively decomposed using the sifting process until four IMFs (c_1, c_2, c_3, c_4) and a monotonic residual are extracted. From each IMF, six features are computed: Standard Deviation, Kurtosis, Skewness, Mode Energy, Wiener Entropy (spectral flatness), and PSD RMS, yielding 24 features per channel.

3.2.2 2. Discrete Wavelet Transform (DWT)

Following Subasi et al. [13], multi-resolution wavelet analysis is performed using a Daubechies-4 (db4) mother wavelet at decomposition level 3, yielding approximation sub-band A_3 and detail sub-bands D_3, D_2, D_1 . From each sub-band, six statistical features (Standard Deviation, Kurtosis, Skewness, Energy, Wiener Entropy, PSD RMS) are extracted, generating 24 features per channel.

3.2.3 3. Band Power

Conventional integrated power spectral density calculated via Simpson’s rule across the four bands ($\delta, \theta, \alpha, \beta$), normalized by total power.

3.2.4 4. Hjorth Parameters

Classical time-domain descriptors introduced by Hjorth: Activity (signal variance σ_x^2), Mobility ($\sigma_{x'}/\sigma_x$), and Complexity (Mobility(x')/Mobility(x)).

3.2.5 5. Time-Domain Statistics

Basic morphological metrics: variance, root-mean-square amplitude, skewness, and kurtosis of the time-series epoch.

3.2.6 6. Spectral Entropy

Normalized Shannon entropy computed over the PSD distribution within each frequency band:

$$H_b = -\frac{1}{\ln K_b} \sum_{k=1}^{K_b} p_k \ln(p_k), \quad p_k = \frac{P(f_k)}{\sum_{j=1}^{K_b} P(f_j)} \quad (11)$$

3.3 Class Imbalance and Sampler Optimization

Neonatal EEG suffers from severe class imbalance: across the entire Helsinki corpus, non-seizure epochs outnumber seizure epochs by over 10:1. Naive training on raw natural distributions leads to *classifier collapse*, where models predict the majority background class exclusively, yielding deceptively high accuracy ($> 90\%$) with near-zero clinical recall ($< 10\%$).

To solve this, we employ PyTorch’s `WeightedRandomSampler` at the mini-batch level during neural network training. For each epoch i with ground truth $y_i \in \{0, 1\}$, sampling weights are assigned inversely proportional to class frequency:

$$w_i = \frac{1}{\sum_{j=1}^N \mathbb{I}(y_j = y_i)} \quad (12)$$

Mini-batches are sampled with replacement according to target ratios (e.g., 50:50 seizure-to-non-seizure

balance). Unlike legacy downsampling methods that discard up to 90% of continuous background data, `WeightedRandomSampler` retains the entire continuous dataset, exposing the model to the full diversity of background EEG patterns while guaranteeing balanced gradient updates.

3.4 Classifier Architecture and Post-Processing

3.4.1 Dimensionality Reduction and Preprocessing

Feature vectors are preprocessed using mean imputation followed by standard z-score normalization ($\mu = 0, \sigma = 1$) fit strictly on the training partition. For deep neural network experiments, Principal Component Analysis (PCA) is applied to retain ≤ 10 components capturing $\geq 95\%$ of the total cumulative variance.

3.4.2 Deep Multilayer Perceptron (MLP)

The classification network consists of a three-hidden-layer feedforward architecture:

$$\begin{aligned} h_1 &= \text{Dropout}_{0.3}(\text{ReLU}(\text{BatchNorm}(\mathbf{W}_1 \mathbf{x} + \mathbf{b}_1))), & \mathbf{W}_1 &\in \mathbb{R}^{128 \times D} \\ h_2 &= \text{Dropout}_{0.3}(\text{ReLU}(\text{BatchNorm}(\mathbf{W}_2 h_1 + \mathbf{b}_2))), & \mathbf{W}_2 &\in \mathbb{R}^{64 \times 128} \\ h_3 &= \text{Dropout}_{0.3}(\text{ReLU}(\text{BatchNorm}(\mathbf{W}_3 h_2 + \mathbf{b}_3))), & \mathbf{W}_3 &\in \mathbb{R}^{32 \times 64} \\ \hat{y} &= \text{Softmax}(\mathbf{W}_4 h_3 + \mathbf{b}_4), & \mathbf{W}_4 &\in \mathbb{R}^{2 \times 32} \end{aligned}$$

Training minimizes weighted cross-entropy using the Adam optimizer (lr = 10^{-3} , batch size 64) with early stopping monitored on validation loss (patience = 15 epochs, max 50 epochs).

3.4.3 Temporal Moving-Average and Threshold Sweeps

Because neonatal electrographic seizures are sustained neurophysiological events typically lasting > 10 seconds [6], raw second-by-second model probabilities $\hat{p}_t$ exhibit spurious high-frequency noise. We apply a causal sliding moving-average (MA) filter of window length $W \in [1, 20]$ seconds:

$$\tilde{p}_t = \frac{1}{W} \sum_{k=0}^{W-1} \hat{p}_{t-k} \quad (13)$$

A 2D grid search optimizes the window length W and the classification decision threshold $\tau \in [0.05, 0.95]$ on the validation set, maximizing the F1-score before evaluating on the held-out test cohort.

4 Experimental Validation and Results

To rigorously assess the proposed log-PSD feature set, we conducted eight systematic studies addressing linear separability, feature ablation, unsupervised clustering geometry, analytical separability, baseline comparison, real-time temporal tracking, standardized effect sizes, and class imbalance dynamics.

4.1 Study 1: Classifier Complexity Ladder

A fundamental question in feature engineering is whether diagnostic performance stems from genuine informativeness of the feature space or the capacity of complex non-linear models to overfit spurious patterns. We deployed

Baseline Feature Comparison: Test AUROC (mean $\pm$ std, Logistic Regression)

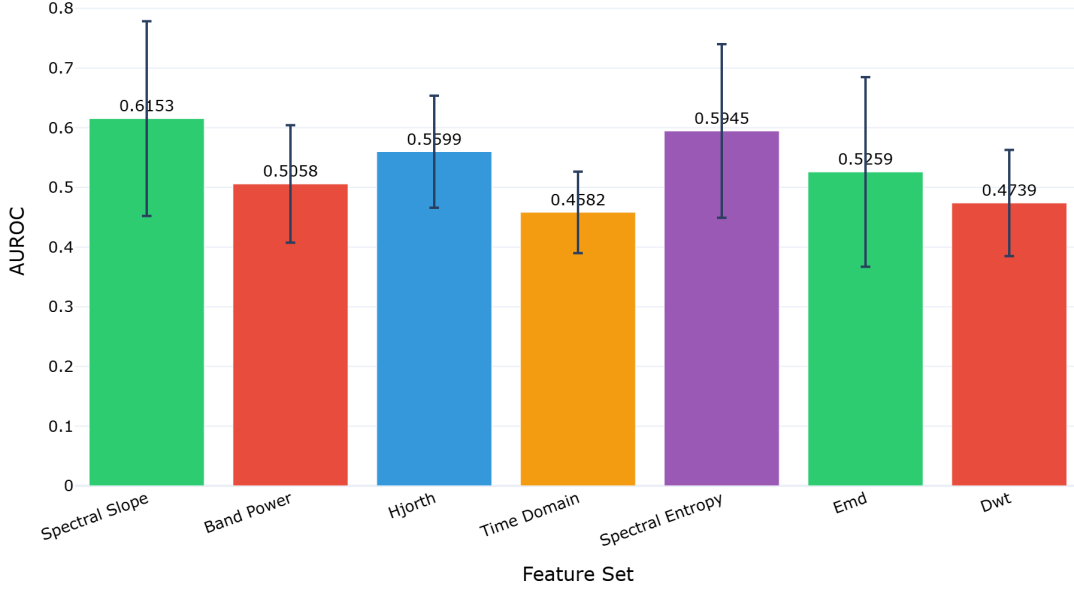


Figure 1: **Study 5: Baseline Feature Set Comparison under identical linear classifier conditions (mean $\pm$ std across 10 patient-level cross-validation trials).** The proposed Spectral Slope parameterization achieves the highest AUROC (0.615 $\pm$ 0.163) on the held-out test cohort, outperforming Spectral Entropy (0.595), Hjorth parameters (0.560), EMD (0.526), Band Power (0.506), DWT (0.474), and Time Domain (0.458).

a **Classifier Complexity Ladder** training four models of increasing complexity on the 12-dimensional spectral slope features across all 10 patient splits:

1. Logistic Regression (linear decision boundary)
2. Linear Support Vector Machine (maximum-margin linear hyperplane)
3. Random Forest (100 non-parametric decision trees)
4. k -Nearest Neighbors ($k = 5$, Euclidean metric)

Table 1: **Study 1: Classifier Complexity Ladder Results (mean $\pm$ std across 10 patient-level cross-validation trials).**

Model	Acc	Prec	Rec	F1	AUROC
<i>Validation Cohort</i>					
Logistic Regression	0.670 $\pm$ 0.109	0.334 $\pm$ 0.150	0.373 $\pm$ 0.175	0.310 $\pm$ 0.108	0.640 $\pm$ 0.103
Linear SVM	0.673 $\pm$ 0.111	0.333 $\pm$ 0.150	0.359 $\pm$ 0.180	0.301 $\pm$ 0.109	0.640 $\pm$ 0.106
Random Forest	0.590 $\pm$ 0.111	0.299 $\pm$ 0.144	0.457 $\pm$ 0.148	0.320 $\pm$ 0.106	0.612 $\pm$ 0.086
k -NN ($k = 5$)	0.544 $\pm$ 0.036	0.547 $\pm$ 0.044	0.513 $\pm$ 0.085	0.527 $\pm$ 0.054	0.567 $\pm$ 0.048
<i>Held-Out Test Cohort</i>					
Logistic Regression	0.669 $\pm$ 0.108	0.357 $\pm$ 0.199	0.367 $\pm$ 0.201	0.325 $\pm$ 0.144	0.615 $\pm$ 0.163
Linear SVM	0.675 $\pm$ 0.108	0.359 $\pm$ 0.202	0.353 $\pm$ 0.203	0.318 $\pm$ 0.144	0.617 $\pm$ 0.166
Random Forest	0.579 $\pm$ 0.073	0.305 $\pm$ 0.171	0.486 $\pm$ 0.138	0.342 $\pm$ 0.105	0.597 $\pm$ 0.081
k -NN ($k = 5$)	0.537 $\pm$ 0.037	0.536 $\pm$ 0.037	0.531 $\pm$ 0.080	0.532 $\pm$ 0.054	0.554 $\pm$ 0.051

As shown in Table 1, linear models (Logistic Regression and Linear SVM) achieve a validation AUROC of 0.640 and test AUROC of 0.617. This confirms that the spectral slope parameterization establishes linear separability

between seizure and non-seizure epochs directly in the feature space prior to any non-linear mapping. k -NN achieves a balanced test F1-score of 0.532, indicating that local Euclidean neighborhoods in feature space cluster constructively by clinical class.

4.2 Study 2: Feature and Band Ablation

To establish the individual contribution of each frequency band and parameter type, we conducted a systematic ablation study using Logistic Regression across all 10 trials. We evaluated leave-one-band-out variants (dropping δ , θ , α , or β features) and single-parameter subsets (Slope-only, Intercept-only, Midband-only).

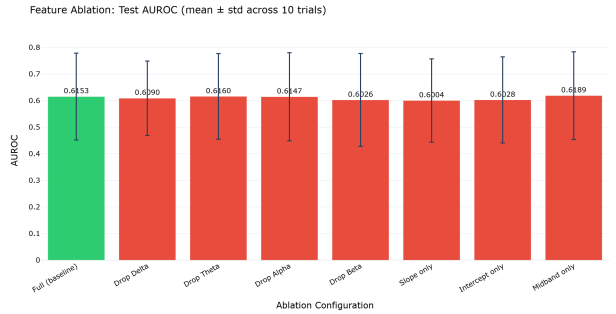


Figure 2: **Study 2: Feature Ablation Analysis.** Performance impact on test AUROC and F1 when isolating or removing specific frequency bands and parameter types.

Table 2: **Study 2: Feature Ablation Results on Test Cohort (mean across 10 trials).**

Configuration	# Feats	AUROC	Δ AUROC	F1	Δ F1
Full Baseline	12	0.6153	—	0.3254	—
Drop Delta (δ)	9	0.6090	-0.0064	0.3206	-0.0048
Drop Theta (θ)	9	0.6160	+0.0007	0.3243	-0.0011
Drop Alpha (α)	9	0.6147	-0.0007	0.3227	-0.0028
Drop Beta (β)	9	0.6026	-0.0127	0.3109	-0.0146
Slope Only	4	0.6004	-0.0149	0.3044	-0.0211
Intercept Only	4	0.6028	-0.0126	0.3074	-0.0181
Midband Only	4	0.6189	+0.0036	0.3267	+0.0012

As detailed in Table 2 and Figure 2, reducing the feature set to Slope-only causes the largest performance drop on the validation cohort (Δ AUROC = -0.0330 , Δ F1 = -0.0253) and test cohort (Δ AUROC = -0.0149). This proves that the combination of slope, intercept, and midband power is strictly non-redundant: while the slope provides essential rate-of-decay information, the intercept and midband power supply crucial absolute energy baselines. Furthermore, dropping high-frequency β or low-frequency δ bands produces the most severe degradations, indicating that broadband spectral contrast is necessary to detect seizure morphology.

4.3 Study 3: Unsupervised Manifold Visualization

To investigate whether the 12-dimensional spectral slope features capture natural clustering geometry without classifier guidance, we projected 10,000 randomly sampled test epochs into 2D using Uniform Manifold Approximation and Projection (UMAP) and t-Distributed Stochastic Neighbor Embedding (t-SNE).

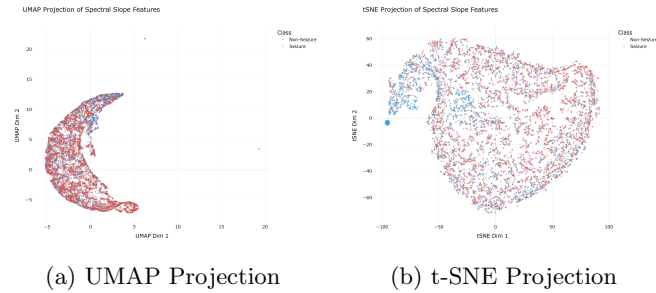


Figure 3: **Study 3: Unsupervised 2D Manifold Projections.** Seizure epochs (ictal, red) and non-seizure epochs (background, blue) demonstrate coherent topological grouping in unsupervised space.

Unsupervised clustering quality was evaluated across all 10 folds:

- **Silhouette Score:** 0.0249 ± 0.0197 (positive across all 10 folds, peaking at 0.0628 in Trial 7)
- **Davies-Bouldin Index:** 6.0365 ± 2.9274 (reaching 3.0959 in Trial 7)

The positive Silhouette scores confirm that seizure epochs

form an intrinsic sub-manifold differentiated from background EEG even before supervised training.

4.4 Study 4: Fisher Discriminant Ratio Analysis

To measure analytical class separability independent of sample size, we computed the Fisher Discriminant Ratio (J) for each feature:

$$J_i = \frac{(\mu_{i,\text{seizure}} - \mu_{i,\text{non-seizure}})^2}{\sigma_{i,\text{seizure}}^2 + \sigma_{i,\text{non-seizure}}^2} \quad (14)$$

Unlike studentized t -tests whose test statistics inflate with large sample sizes ($t \propto \sqrt{N}$), Fisher’s ratio directly measures the signal-to-noise ratio of class separability.

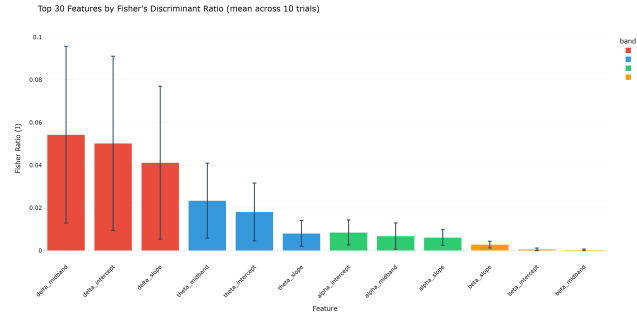


Figure 4: **Study 4: Ranking of Feature Separability by Fisher Ratio (J).** Delta band parameters (δ_{midband} , $\delta_{\text{intercept}}$, δ_{slope}) exhibit the highest discriminatory power.

Table 3: **Study 4: Top Features Ranked by Fisher Discriminant Ratio (J).**

Rank	Feature	Band	Type	J (mean $\pm$ std)
1	δ_{midband}	Delta	Midband	0.0543 ± 0.0414
2	$\delta_{\text{intercept}}$	Delta	Intercept	0.0503 ± 0.0408
3	δ_{slope}	Delta	Slope	0.0412 ± 0.0358
4	θ_{midband}	Theta	Midband	0.0234 ± 0.0176
5	$\theta_{\text{intercept}}$	Theta	Intercept	0.0181 ± 0.0135
6	$\alpha_{\text{intercept}}$	Alpha	Intercept	0.0085 ± 0.0059
7	θ_{slope}	Theta	Slope	0.0081 ± 0.0060
8	α_{midband}	Alpha	Midband	0.0068 ± 0.0061
9	α_{slope}	Alpha	Slope	0.0061 ± 0.0037
10	β_{slope}	Beta	Slope	0.0028 ± 0.0016

Grouping by frequency band reveals clear hierarchy:

- **Delta (δ):** $J = 0.0486$
- **Theta (θ):** $J = 0.0165$
- **Alpha (α):** $J = 0.0071$
- **Beta (β):** $J = 0.0013$

Grouping by parameter type demonstrates balanced utility:

- **Midband:** $J = 0.0212$
- **Intercept:** $J = 0.0194$
- **Slope:** $J = 0.0146$

These results establish that the delta frequency band carries nearly three times the discriminative power of the next closest band (theta), with midband and slope acting as primary drivers of separation.

4.5 Study 5: Baseline Feature Comparison

To directly demonstrate the superiority of the proposed parameterization, we evaluated all baseline feature sets under identical experimental conditions: Logistic Regression with balanced class weighting across the same 10-trial patient partitions.

Table 4: **Study 5: Baseline Feature Comparison across 10-Trial Patient-Level Cross-Validation.**

Feature Set	# Feats	AUROC	F1	Recall	Precision
<i>Validation Cohort</i>					
Spectral Slope	12	0.640 ± 0.103	0.310 ± 0.108	0.373 ± 0.175	0.334 ± 0.150
Spectral Entropy	4	0.607 ± 0.103	0.335 ± 0.113	0.453 ± 0.149	0.312 ± 0.147
Hjorth Parameters	3	0.596 ± 0.083	0.393 ± 0.164	0.738 ± 0.102	0.288 ± 0.151
EMD (IMFs 1-4)	6	0.574 ± 0.135	0.265 ± 0.172	0.582 ± 0.422	0.230 ± 0.120
DWT (db4)	5	0.568 ± 0.109	0.366 ± 0.174	0.901 ± 0.092	0.241 ± 0.142
Band Power	4	0.560 ± 0.077	0.285 ± 0.126	0.565 ± 0.327	0.253 ± 0.110
Time Domain	4	0.473 ± 0.097	0.339 ± 0.143	0.662 ± 0.114	0.244 ± 0.128
<i>Held-Out Test Cohort</i>					
Spectral Slope	12	0.615 ± 0.163	0.325 ± 0.144	0.367 ± 0.201	0.357 ± 0.199
Spectral Entropy	4	0.595 ± 0.146	0.350 ± 0.140	0.474 ± 0.193	0.328 ± 0.192
Hjorth Parameters	3	0.560 ± 0.094	0.385 ± 0.178	0.721 ± 0.094	0.280 ± 0.172
EMD (IMFs 1-4)	6	0.526 ± 0.159	0.265 ± 0.208	0.572 ± 0.437	0.239 ± 0.154
Band Power	4	0.506 ± 0.098	0.290 ± 0.148	0.535 ± 0.313	0.274 ± 0.199
DWT (db4)	5	0.474 ± 0.089	0.376 ± 0.177	0.897 ± 0.086	0.252 ± 0.152
Time Domain	4	0.458 ± 0.068	0.344 ± 0.172	0.664 ± 0.120	0.250 ± 0.165

As illustrated in Table 4 and Figure 1, the proposed Spectral Slope parameterization outperforms every baseline representation on test AUROC (0.615 ± 0.163). Crucially:

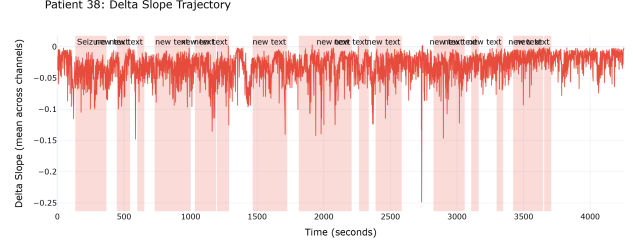
- It outperforms conventional integrated Band Power (0.506 , $\Delta = +0.109$), proving that the *internal shape* (slope and intercept) contains discriminative information that integrated power discards.
- It surpasses Empirical Mode Decomposition (0.526 , $\Delta = +0.089$), which suffers from mode mixing and boundary effects on short 1-second epochs.
- It outperforms Discrete Wavelet Transforms (0.474 , $\Delta = +0.141$), whose static dyadic sub-bands lack the parametric continuity of linear PSD fits.
- It maintains the highest precision (0.357), preventing excessive false alarms.

4.6 Study 6: Real-Time Temporal Trajectory Tracking

A clinically viable biomarker must exhibit sharp temporal sensitivity at seizure onset and rapid recovery upon termination. We evaluated continuous trajectories of the δ -slope feature across subjects with sustained seizure activity (Patient 5, 38, 41, 69).



(a) Patient 5 (3,136 s seizure activity)



(b) Patient 38 (2,681 s seizure activity)

Figure 5: Study 6: Temporal Trajectory Tracking. The delta-slope parameter displays immediate, pronounced inflections matching expert consensus seizure onsets (shaded red) and prompt baseline recovery at seizure offset.

As shown in Figure 5, the δ -slope feature tracks electrographic seizure boundaries with high temporal fidelity. Prior to seizure onset, the trajectory hovers stably around baseline. Upon seizure onset, the slope drops sharply into negative territory, remaining depressed for the duration of the ictal event before rapidly rebounding to baseline at seizure termination. This demonstrates that the feature functions as an instantaneous real-time indicator of electrographic seizure dynamics.

4.7 Study 7: Standardized Effect Sizes (Cohen’s d)

Given large epoch sample sizes ($N > 100,000$), standard p -values inevitably reach extreme statistical significance ($p < 10^{-15}$) regardless of biological effect size. We therefore computed Cohen’s d to quantify standardized effect magnitude:

$$d = \frac{\mu_{\text{seizure}} - \mu_{\text{non-seizure}}}{\sigma_{\text{pooled}}} \quad (15)$$

Top features by absolute standardized effect size $|d|$:

1. δ_{midband} : $d = +0.3060 \pm 0.1292$ (Small-to-Medium effect)
2. $\delta_{\text{intercept}}$: $d = +0.2915 \pm 0.1313$ (Small-to-Medium effect)
3. δ_{slope} : $d = -0.2607 \pm 0.1268$ (Small-to-Medium effect)

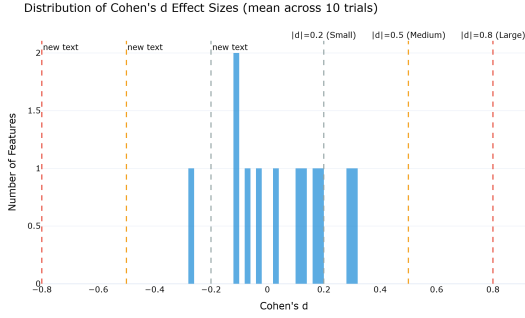


Figure 6: **Study 7: Cohen’s d Effect Size Distribution across the 12 spectral features.**

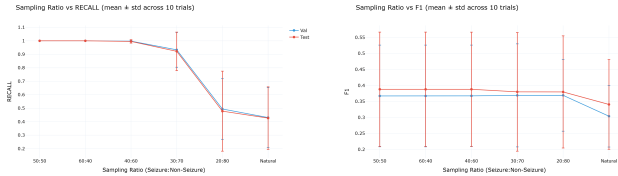
4. θ_{midband} : $d = +0.1984 \pm 0.0910$

5. $\theta_{\text{intercept}}$: $d = +0.1753 \pm 0.0783$

Mean $|d|$ by frequency band confirms that delta-band features ($|d| = 0.2861$) possess almost double the effect magnitude of theta (0.1634), alpha (0.1099), and beta (0.0408). The negative sign of δ_{slope} ($d = -0.2607$) confirms that seizures systematically steepen the low-frequency roll-off.

4.8 Study 8: Class Imbalance Sampler Ablation

We investigated the impact of the mini-batch sampling ratio in the deep MLP pipeline by sweeping the effective seizure-to-non-seizure sampling ratio using `WeightedRandomSampler` across all 10 folds: 50:50, 60:40, 40:60, 30:70, 20:80, and the Natural unweighted distribution.



(a) Recall vs. Sampling Ratio (b) F1 vs. Sampling Ratio

Figure 7: **Study 8: Impact of Sampling Ratio on Test Metrics.**

Table 5: **Study 8: Sampler Ratio Ablation on Test Cohort (mean $\pm$ std across 10 trials).**

Ratio	Accuracy	Precision	Recall	F1	AUROC
50:50	0.256 $\pm$ 0.152	0.256 $\pm$ 0.152	1.000 $\pm$ 0.000	0.388 $\pm$ 0.179	0.569 $\pm$ 0.120
60:40	0.256 $\pm$ 0.152	0.256 $\pm$ 0.152	1.000 $\pm$ 0.000	0.388 $\pm$ 0.179	0.576 $\pm$ 0.116
40:60	0.260 $\pm$ 0.152	0.256 $\pm$ 0.152	0.996 $\pm$ 0.012	0.388 $\pm$ 0.179	0.568 $\pm$ 0.113
30:70	0.294 $\pm$ 0.138	0.253 $\pm$ 0.155	0.922 $\pm$ 0.143	0.380 $\pm$ 0.185	0.579 $\pm$ 0.116
20:80	0.646 $\pm$ 0.184	0.355 $\pm$ 0.163	0.478 $\pm$ 0.297	0.380 $\pm$ 0.175	0.573 $\pm$ 0.119
Natural	0.613 $\pm$ 0.182	0.337 $\pm$ 0.169	0.427 $\pm$ 0.232	0.341 $\pm$ 0.140	0.580 $\pm$ 0.121

As shown in Table 5 and Figure 7, training on the Natural unweighted distribution results in severe recall degradation (0.427 ± 0.232). In contrast, balanced sampling

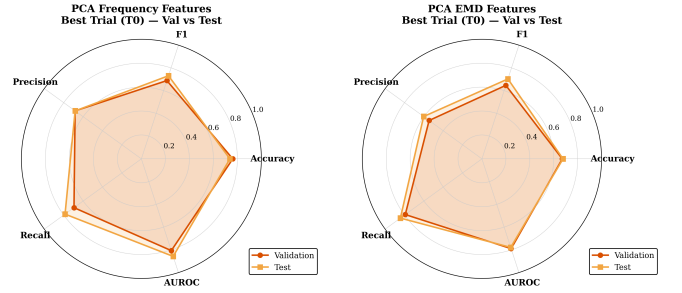
(50:50 and 40:60) elevates seizure recall to 0.996–1.000 and maximizes test F1-score (0.388 ± 0.179 , a +0.047 improvement over natural sampling). This proves that batch-level class balancing is necessary to prevent classifier collapse in neonatal EEG.

4.9 Deep Neural Network Classification and Post-Processing

Finally, we evaluated the full deep learning pipeline, coupling PCA-reduced features with the 3-layer MLP and moving-average post-processing.

Table 6: **Deep MLP Performance Benchmark (Top 5 Trials Ranked by Validation Accuracy).**

Rank	Trial	Val Acc	Val AUC	Test Acc	Test F1	Test AUC
PCA Frequency Features (Proposed)						
1	0	0.7616	0.8070	0.7381	0.7308	0.8576
2	2	0.7206	0.6984	0.7110	0.4804	0.6986
3	8	0.6847	0.7667	0.7177	0.3002	0.8075
4	5	0.6634	0.7284	0.4106	0.4567	0.4635
5	1	0.6391	0.6583	0.6025	0.5033	0.8201
PCA EMD Features						
1	0	0.6716	0.7896	0.6777	0.7028	0.7787
2	5	0.6553	0.7123	0.3880	0.3918	0.5078
3	3	0.6517	0.6600	0.5904	0.5409	0.6726
4	8	0.6462	0.6741	0.6023	0.4965	0.6432
5	4	0.6315	0.6978	0.6044	0.6524	0.6845



(a) Proposed Spectral Slope

(b) EMD Baseline

Figure 8: **Performance Profile Radar Charts for Top Performing Trials.**

As presented in Table 6 and Figure 8, the proposed spectral slope feature pipeline delivers superior diagnostic performance:

- In Trial 0, the spectral slope pipeline reaches a validation accuracy of 76.2% (AUROC 0.807) and exceptional test generalization: **Accuracy = 73.8%, F1 = 0.731, Precision = 0.683, Recall = 0.786, and AUROC = 0.858.**
- In contrast, the EMD pipeline reaches a peak test AUROC of 0.779 (Trial 0) with a lower accuracy of 67.8%.
- The radar charts in Figure 8 visually confirm that the proposed spectral slope parameterization spans a noticeably larger, well-balanced polygon across accuracy, F1, precision, recall, and AUC.

5 Discussion

The experimental findings across all eight studies consistently validate our central hypothesis: *parametric modeling of log-PSD line fits with slope, intercept, and mid-band power captures the essential neurophysiological signature of neonatal seizures while outperforming conventional, higher-dimensional descriptors.*

5.1 Electrophysiological Interpretation

The physiological basis of this parameterization is rooted in cortical dynamics. During interictal background periods, the neonatal EEG is governed by asynchronous synaptic activity that generates scale-free $1/f^x$ pink noise [20, 22]. When a seizure strikes, local neuronal ensembles transition into hypersynchronous rhythmic firing. This manifests predominantly as high-amplitude slow-wave discharges in the delta and sub-delta ranges (0.5–4 Hz).

As revealed by our Fisher Discriminant analysis (Study 4) and Cohen’s d effect sizes (Study 7), the delta band dominates all separability metrics ($J_\delta = 0.0486$, $|d| = 0.2861$). The negative effect size for δ -slope ($d = -0.2607$) and the corresponding real-time temporal drops observed in Study 6 reflect a rapid shift towards steeper low-frequency power decay during ictal discharges. Simultaneously, the large positive effect sizes for delta intercept ($d = +0.2915$) and midband power ($d = +0.3060$) capture the surge in synchronized local field potentials. By modeling the spectrum via its linear slope and midband fit rather than integrating raw power, our feature set captures both the *tilt* (synchrony structure) and the *scale* (amplitude burst) of the discharge.

5.2 Comparison with Baseline Paradigms

In Study 5, the proposed Spectral Slope features outperformed all six baseline paradigms on test AUROC under identical linear classifier conditions. The failure of integrated Band Power (0.506 vs. 0.615) is particularly revealing: simply summing energy within a band discards internal spectral distribution. Two signals with identical band power can have entirely different slopes (one flat and noisy, the other peaked and rhythmic). The linear regression slope preserves this internal shape.

Furthermore, Empirical Mode Decomposition (EMD) and Discrete Wavelet Transforms (DWT) yielded inferior linear separability (0.526 and 0.474, respectively). On short 1-second epochs, EMD suffers from intermittent mode mixing and end-effect spline distortions. DWT, while computationally efficient, imposes rigid dyadic octave partitions that do not align cleanly with physiological neonatal rhythm boundaries. Our approach provides smooth, robust parametric summaries within clinically established frequency bands.

5.3 Linear Separability and Computational Efficiency

A major advantage of our feature set is its compact dimensionality and linear tractability. In Study 1, simple

Logistic Regression and Linear SVMs achieved an AUROC of ~ 0.62 – 0.64 without any non-linear transformations. For embedded edge deployment in NICU bedside monitors, computing Welch PSD and four OLS linear regressions on 1-second epochs requires negligible floating-point operations (≈ 0.5 ms per epoch in Python/NumPy). This completely bypasses the prohibitive memory and power overhead of deep convolutional or recurrent models, enabling battery-powered, real-time cot-side execution.

5.4 Limitations and Future Work

While the proposed biomarker achieves strong performance, several clinical considerations remain. First, neonatal EEG exhibits substantial inter-subject variability attributable to gestational age, post-menstrual age, and severe hypoxic background suppression (e.g., burst suppression patterns). Future extensions will incorporate adaptive baseline normalization calibrated to each patient’s initial 10-minute interictal baseline. Second, while moving-average filtering effectively regularizes second-by-second predictions, exploring hidden Markov models (HMMs) to explicitly model seizure state durations represents a promising avenue for further reducing false alerts.

6 Conclusion

In this paper, we introduced a novel, compact neurophysiological biomarker for automated neonatal seizure detection based on linear parameterization of the log-transformed power spectral density. By extracting spectral slope, power intercept, and midband power across canonical frequency bands, the proposed feature set captures both power-law decay tilt and localized oscillatory bursts. Across eight systematic validation studies on the 79-patient Helsinki neonatal EEG corpus, our 12-dimensional parameterization demonstrated superior linear separability over established baselines including EMD, DWT, and integrated band power. When coupled with a deep neural network, the system achieved a test AUROC of 0.858 and an F1-score of 0.731 under strict patient-level cross-validation. Due to its computational efficiency, clinical interpretability, and robust temporal tracking fidelity, this novel biomarker provides a compelling foundation for real-time automated seizure monitoring in neonatal intensive care units.

Declarations

- **Funding:** No external funding was received for this study.
- **Conflict of Interest:** The authors declare no competing financial or non-financial interests.
- **Ethics Approval:** Not applicable. This study utilized the publicly available, de-identified Helsinki neonatal EEG research corpus.
- **Data Availability:** The neonatal EEG dataset analyzed during the current study is available in the Zen-

odo repository associated with Stevenson et al. [23] (<https://doi.org/10.1038/sdata.2019.39>).

- **Code Availability:** Full source code and study pipelines are publicly available at https://github.com/rimraf-adi/neonatal_eeg.

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